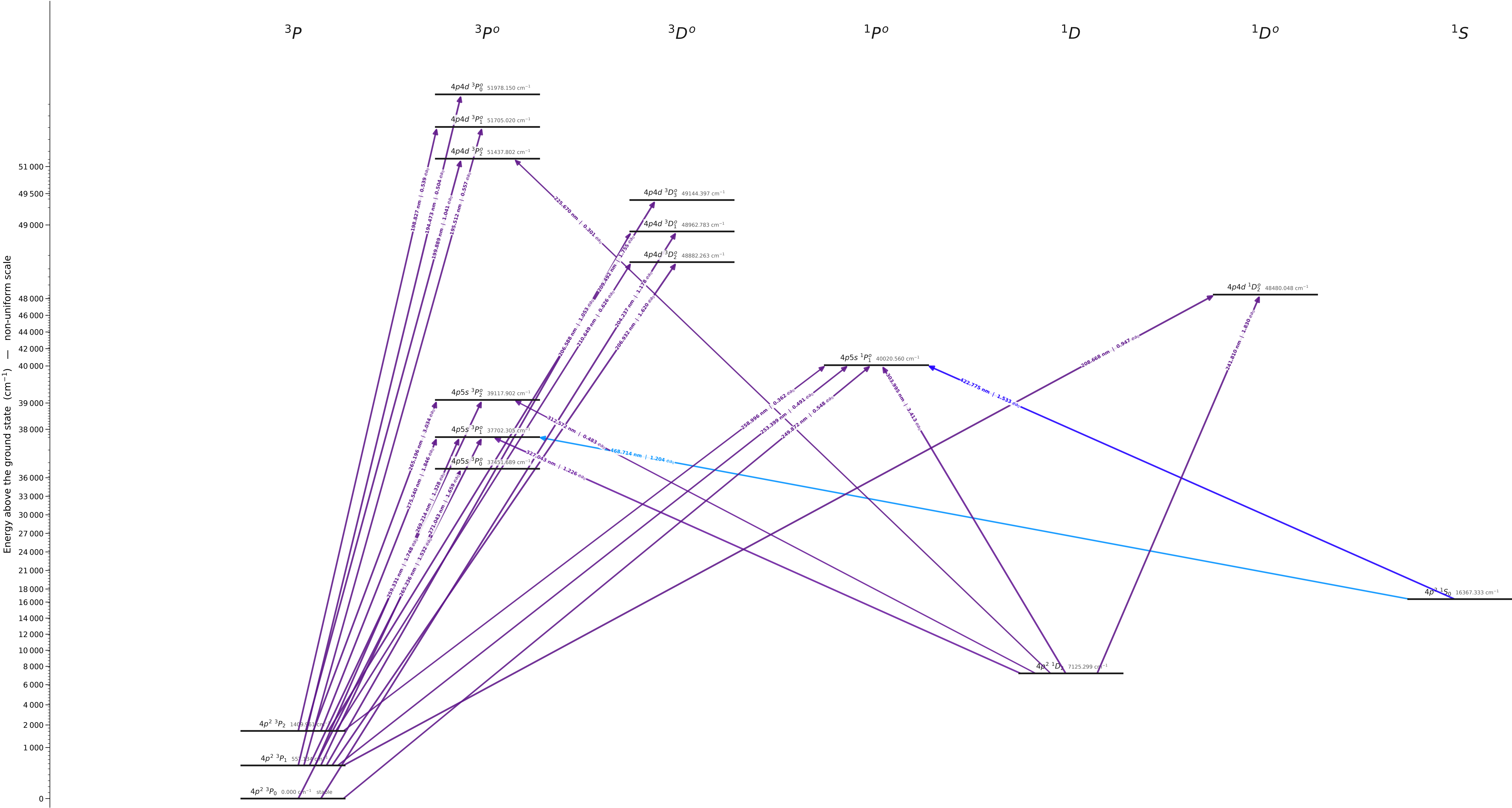
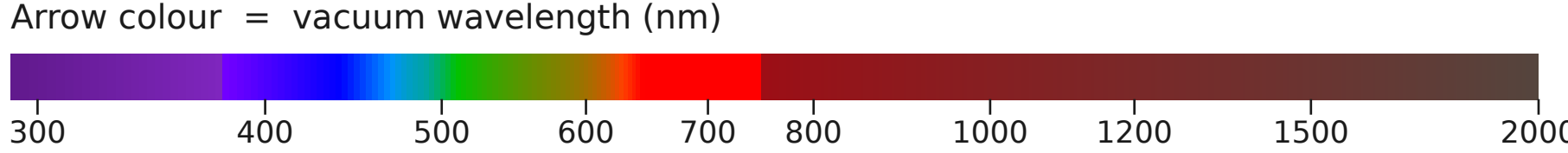


Germanium

Ge (Ge I). Strongest classified transitions below 2 μm (NIST): 16 levels, 26 lines.

Ge⁺ ionisation limit 63713.24 cm⁻¹ = 7.89943 eV (156.953 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J||er||J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	(J er J') (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
4p ² ³ P ₁ – 4p4d ³ P ₀ ^o	194.4730	–	1541.5633	0.504	7.000e+07	–	–	–
4p ² ³ P ₂ – 4p4d ³ P ₂ ^o	199.8887	–	1499.7969	1.041	5.500e+07	–	–	–
4p ² ³ P ₀ – 4p4d ³ D ₁ ^o	204.2368	204.1712	1467.8673	1.178	1.100e+08	–	–	–
4p ² ³ P ₁ – 4p4d ³ D ₁ ^o	206.5875	206.5215	1451.1648	1.053	8.500e+07	–	–	–
4p ² ³ P ₁ – 4p4d ³ D ₂ ^o	206.9317	206.8656	1448.7509	1.620	1.200e+08	–	–	–
4p ² ³ P ₁ – 4p4d ¹ D ₂ ^o	208.6684	208.6021	1436.6928	0.947	4.000e+07	–	–	–
4p ² ³ P ₂ – 4p4d ³ D ₂ ^o	209.4924	209.4258	1431.0424	1.755	9.700e+07	–	–	–
4p ² ¹ D ₂ – 4p4d ¹ D ₂ ^o	241.8102	241.7367	1239.7842	1.830	9.600e+07	–	–	–
4p ² ³ P ₁ – 4p5s ³ P ₂ ^o	259.3309	259.2534	1156.0227	1.748	7.100e+07	–	–	–
4p ² ³ P ₂ – 4p5s ³ P ₂ ^o	265.1961	265.1172	1130.4556	3.034	2.000e+08	–	–	–
4p ² ³ P ₀ – 4p5s ³ P ₀ ^o	265.2358	265.1568	1130.2867	1.532	8.500e+07	–	–	–
4p ² ³ P ₁ – 4p5s ³ P ₁ ^o	269.2140	269.1341	1113.5842	1.328	6.100e+07	–	–	–
4p ² ³ P ₁ – 4p5s ³ P ₀ ^o	271.0427	270.9624	1106.0709	1.659	2.800e+08	–	–	–
4p ² ³ P ₂ – 4p5s ³ P ₀ ^o	275.5402	275.4588	1088.0171	1.846	1.100e+08	–	–	–
4p ² ¹ D ₂ – 4p5s ¹ P ₁ ^o	303.9952	303.9067	986.1751	3.413	2.800e+08	–	–	–
4p ² ¹ D ₂ – 4p5s ³ P ₁ ^o	327.0431	326.9489	916.6756	1.226	2.900e+07	–	–	–

λ, λ_{air} from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).